

TONKHIL, Mongolia

WMO: 442660

Lat: 46.3128N

Lon: 93.9017E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.7	-17.1	-28.5	1.5	-14.4	-25.2	1.8	-12.9	35.6	5.1	30.4	11.9	4.2	0	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	75.6	54.7	73.0	53.6	70.4	52.7	59.0	71.1	56.9	69.0	55.0	66.7	11.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.0	81.1	66.6	51.5	73.9	64.1	49.4	68.4	62.0	29.2	71.4	27.6	69.2	26.2	67.1	67.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
32.4	27.5	23.7	-24.2	81.3	5.7	3.0	-28.3	83.4	-31.6	85.2	-34.8	86.9	-39.0	89.0
			-24.4	60.9	5.2	3.7	-28.2	63.5	-31.2	65.7	-34.2	67.8	-38.0	70.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	2.9	8.4	22.0	35.4	44.6	55.8	60.1	56.8	46.3	33.1	19.5	7.6		
	DBStd	21.41	11.09	10.59	10.46	9.67	8.17	6.26	5.14	5.85	7.41	8.41	9.06	9.99		
	HDD50	7077	1461	1165	868	440	203	22	2	13	151	523	914	1315		
	HDD65	11763	1926	1585	1333	887	634	280	166	260	560	988	1364	1780		
	CDD50	811	0	0	0	3	35	195	314	223	41	0	0	0		
	CDD65	21	0	0	0	0	0	3	13	5	0	0	0	0		
	CDH74	159	0	0	0	0	0	25	96	38	0	0	0	0		
	CDH80	8	0	0	0	0	0	1	5	2	0	0	0	0		
	WSAvg	9.5	6.8	8.2	10.6	11.6	12.4	9.9	8.9	9.5	10.0	9.7	8.7	7.1		
	PrecAvg	4.1	0.0	0.0	0.1	0.1	0.4	1.1	1.1	0.8	0.2	0.0	0.0	0.0		
Precipitation	PrecMax	7.4	0.2	0.2	0.2	0.3	1.1	3.9	3.2	2.7	1.0	0.2	0.2	0.2		
	PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.5	0.0	0.0	0.0	0.1	0.3	0.8	0.7	0.7	0.2	0.0	0.0	0.0		
	PrecStd	1.5	0.0	0.0	0.0	0.1	0.3	0.8	0.7	0.7	0.2	0.0	0.0	0.0		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	35.8	49.7	62.2	69.8	76.9	80.4	77.5	70.0	57.1	46.4	34.2		
		MCWB	23.5	27.2	36.3	46.4	50.1	56.6	56.8	56.7	49.4	42.3	35.3	26.4		
	2%	DB	26.2	31.1	45.4	57.3	65.7	73.7	76.7	74.5	65.5	52.7	40.3	28.3		
		MCWB	20.1	23.8	33.1	42.0	47.5	53.7	55.7	53.2	46.3	39.3	31.8	22.2		
	5%	DB	22.2	27.4	41.5	53.7	62.5	70.7	73.7	71.6	62.5	49.7	36.0	24.9		
		MCWB	17.3	21.3	30.8	39.5	46.2	52.9	54.3	52.2	45.9	38.0	28.5	19.4		
	10%	DB	18.3	23.6	38.0	50.3	59.2	67.7	71.0	68.5	59.2	46.3	32.3	21.6		
		MCWB	14.2	18.6	29.0	37.8	44.0	51.6	53.8	51.7	44.8	35.9	25.7	17.0		
	0.4%	WB	24.7	28.1	38.4	47.8	53.3	60.1	62.3	61.2	52.1	44.4	35.7	26.9		
		MCDB	29.5	35.3	48.5	60.1	68.0	74.6	74.1	74.7	64.4	54.3	43.6	32.9		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	20.6	24.2	33.6	43.8	49.7	57.2	59.9	57.3	50.0	41.4	32.1	22.6		
		MCDB	26.0	30.2	43.2	55.3	62.8	71.1	71.5	69.8	61.3	50.6	39.8	28.0		
	5%	WB	17.4	21.4	31.3	40.9	47.3	55.1	57.8	55.1	47.9	38.8	28.9	19.6		
		MCDB	22.0	27.4	40.9	51.2	60.1	67.5	69.2	67.1	58.8	47.9	35.8	24.5		
	10%	WB	14.1	18.6	29.0	38.4	45.1	53.2	55.8	53.0	45.9	36.1	25.9	17.0		
		MCDB	18.4	23.4	37.8	49.0	57.5	64.2	67.0	64.9	57.1	45.5	31.9	21.5		
	5% DB	MDBR	17.4	19.5	20.5	19.1	18.5	17.3	16.6	17.6	18.9	18.6	17.9	16.9		
		MCDBR	20.5	22.2	22.8	21.8	22.1	21.3	20.7	22.0	22.6	21.4	20.8	19.5		
		MCWBR	16.6	17.5	15.4	13.3	12.8	11.1	9.8	10.5	12.3	13.6	15.3	15.3		
		MCWBR	20.5	21.6	21.8	20.7	20.8	19.1	17.6	19.0	20.1	20.7	20.2	19.1		
Clear-Sky Solar Irradiance	taub		0.221	0.251	0.290	0.328	0.332	0.341	0.338	0.317	0.286	0.268	0.240	0.224		
		taud	2.422	2.353	2.296	2.259	2.311	2.363	2.401	2.441	2.488	2.468	2.450	2.410		
	Ebn at Noon		292	298	298	294	296	292	292	295	298	288	279	276		
		Edh at Noon	21	28	35	40	39	38	36	33	29	25	20	19		
All-Sky Solar Radiation	RadAvg		605	954	1417	1859	2170	2192	2125	1908	1610	1062	653	488		
	RadStd		35	47	74	92	144	117	109	79	64	61	51	27		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (3 neighbors)		N/A	N/A	-5.90	N/A	-4.07	-4.34	N/A	N/A	N/A	N/A

Nomenclature: See separate page